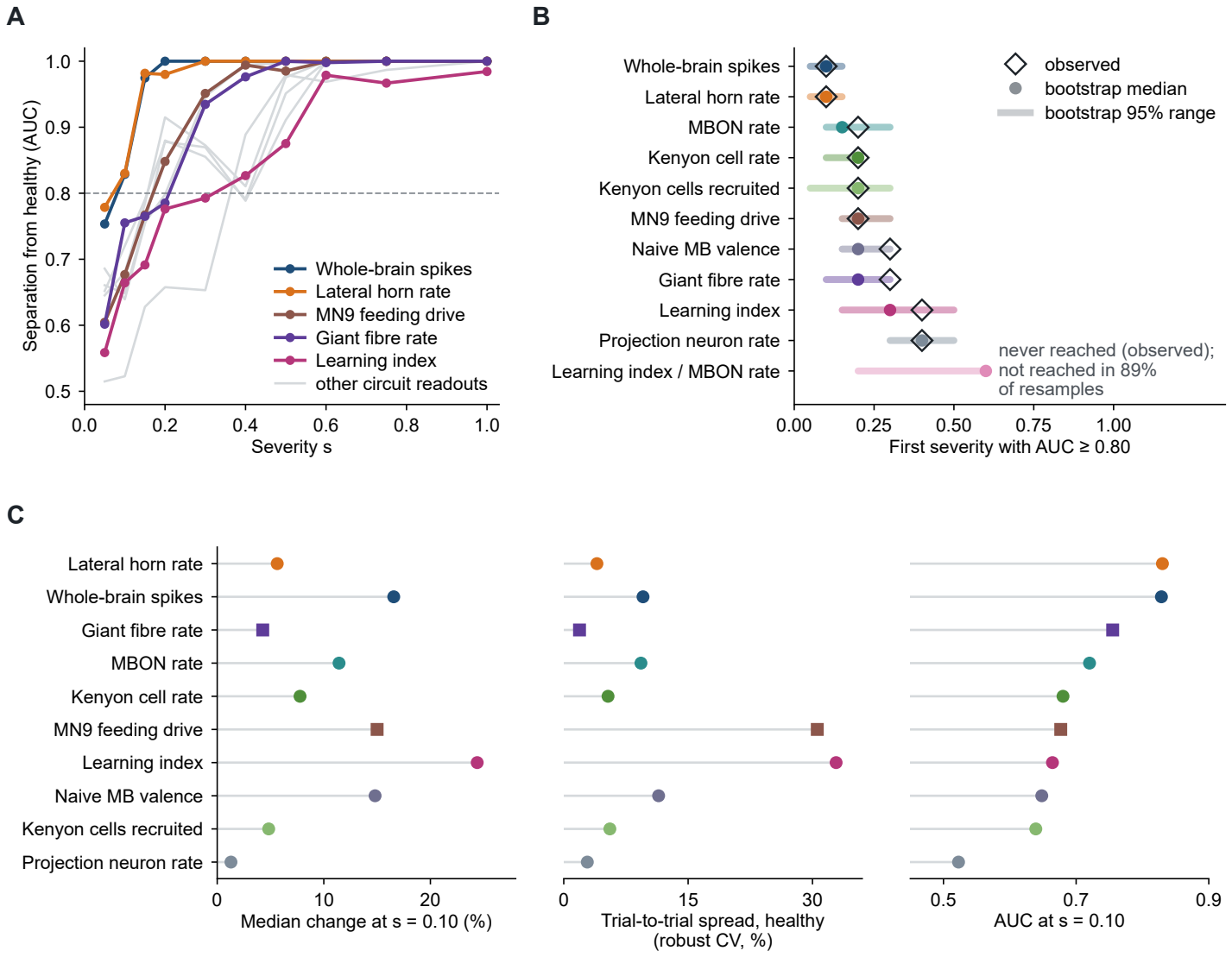

A
Separation from healthy (AUC)
1.0
0.9
0.8
0.7
0.6
0.5
0.0
0.2
0.4
0.6
0.8
1.0
Severity s
Whole-brain spikes
Lateral horn rate
MN9 feeding drive
Giant fibre rate
Learning index
other circuit readouts
B
Whole-brain spikes
Lateral horn rate
MBON rate
Kenyon cell rate
Kenyon cells recruited
MN9 feeding drive
Naive MB valence
Giant fibre rate
Learning index
Projection neuron rate
Learning index / MBON rate
observed
bootstrap median
bootstrap 95% range
never reached (observed);
not reached in 89%
of resamples
0.00
0.25
0.50
0.75
1.00
First severity with AUC ≥ 0.80
C
Lateral horn rate
Whole-brain spikes
Giant fibre rate
MBON rate
Kenyon cell rate
MN9 feeding drive
Learning index
Naive MB valence
Kenyon cells recruited
Projection neuron rate
0
10
20
Median change at s = 0.10 (%)
0
15
30
Trial-to-trial spread, healthy
(robust CV, %)
0.5
0.7
0.9
AUC at s = 0.10